

HANDS-ON SUBNETTING PRACTICAL

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.



Given Network Address: 192.168.10.0/24

To divide this network into 4 equal subnets, we need to borrow 2 bits from the host portion.

New Subnet Mask:

/24 + 2 bits = /26

Subnet Mask in Decimal:

255.255.255.192

Each subnet will have 64 IP addresses (62 usable host addresses).

Subnet No.	Network Address	Usable IP Range	Broadcast Address	Subnet Mask
Subnet 1	192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63	255.255.255.192
Subnet 2	192.168.10.64/26	192.168.10.65 - 192.168.10.126	192.168.10.127	255.255.255.192
Subnet 3	192.168.10.128/26	192.168.10.129 - 192.168.10.190	192.168.10.191	255.255.255.192
Subnet 4	192.168.10.192/26	192.168.10.193 - 192.168.10.254	192.168.10.255	255.255.255.192

Calculation:

- Original Network: 192.168.10.0/24
- Required Subnets: 4
- Borrowed Bits: 2 bits ($2^2 = 4$ subnets)
- New Prefix: /26
- Block Size: $256 - 192 = 64$

Therefore, the four equal subnets are:

1. 192.168.10.0/26
2. 192.168.10.64/26
3. 192.168.10.128/26
4. 192.168.10.192/26

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.
Hint: Make sure not to use the network or broadcast address for any device. ○

To divide the 192.168.10.0/24 network into 4 equal subnets.

Given

- **Network Address: 192.168.10.0/24**
- **Required Subnets: 4**

Step 1: Calculate the Subnet Mask

Number of required subnets = 4

Borrowed bits = 2

New Prefix = /26

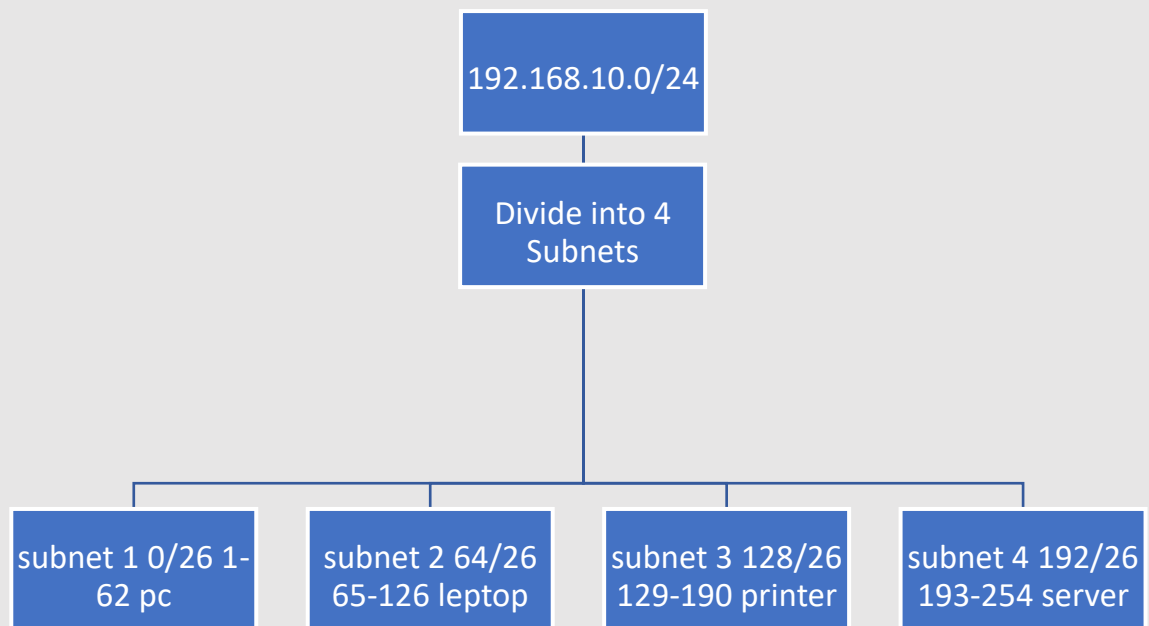
Subnet Mask = 255.255.255.192

Subnet Details

Subnet No.	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192 (/26)	192.168.10.0	192.168.10.63
2	255.255.255.192 (/26)	192.168.10.64	192.168.10.127
3	255.255.255.192 (/26)	192.168.10.128	192.168.10.191
4	255.255.255.192 (/26)	192.168.10.192	192.168.10.255

- 3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.**





4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

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Step 1: Start with the Given Network

The given network address is 192.168.10.0/24. The /24 means the first 24 bits are used for the network part, and the remaining bits are available for host addresses.

Step 2: Divide the Network into 4 Equal Subnets

We need 4 equal subnets.

Using the formula:

$$2^2 = 4$$

So, I borrowed 2 bits from the host portion. New prefix:

$$24 + 2 = /26$$

Therefore, the new subnet mask becomes:

255.255.255.192 (/26)

Step 3: Calculate the Number of Hosts A

/26 subnet leaves 6 host bits.

Total IP addresses in each subnet:

$$2^6 = 64$$

Out of these, the first address is the Network Address and the last address is the Broadcast Address.

So,

Usable Host Addresses = $64 - 2 = 62$ Each subnet can support 62 devices.

Step 4: Create the Four Subnets

Since the block size is 64, the subnet addresses increase by 64 each time.

- Subnet 1 → 192.168.10.0
- Subnet 2 → 192.168.10.64
- Subnet 3 → 192.168.10.128
- Subnet 4 → 192.168.10.192

Step 5: Assign IP Addresses

I assigned one sample device in each subnet.

Device	Assigned IP Address
PC	192.168.10.10
Laptop	192.168.10.70
Printer	192.168.10.140
Server	192.168.10.200

Each IP address belongs to its respective subnet and does not use the network or broadcast address.